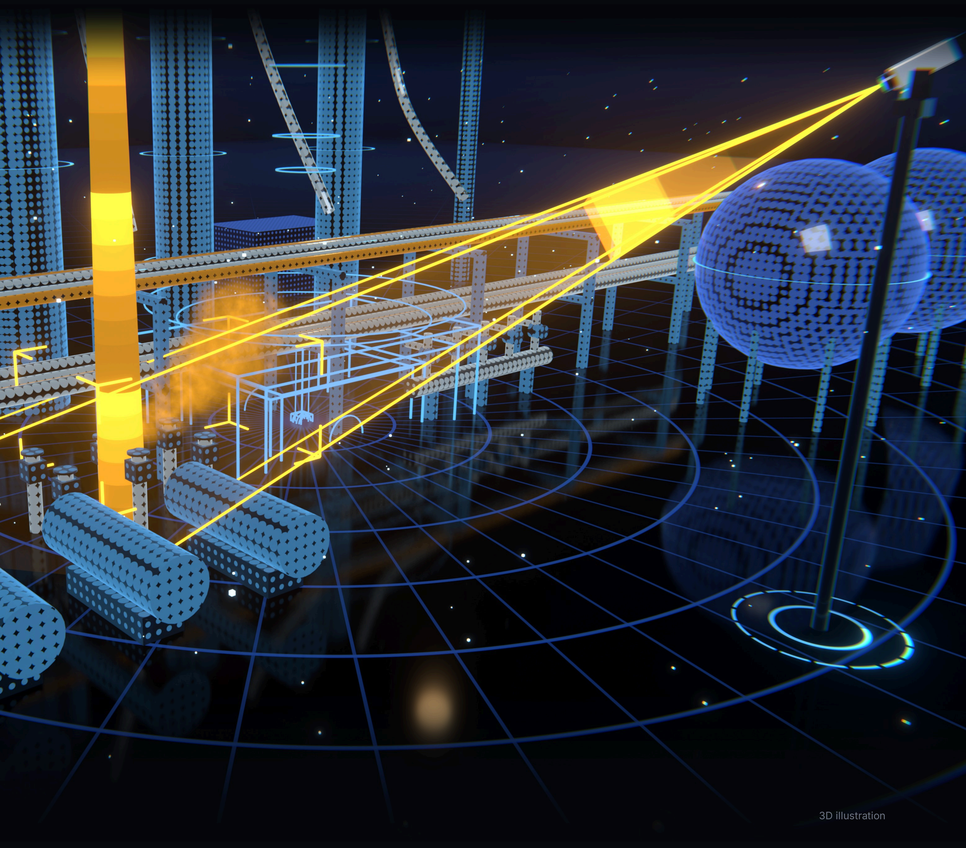
GLD NEXT · DEVELOPMENT CONCEPT

# Optical Gas Imaging

Seeing invisible hydrocarbon leaks, from point to area.



3D illustration

THE PRINCIPLE

## Invisible gas. Made visible.

A cooled infrared camera captures C–H bond absorption near 3.3 µm, so the gas appears as contrast against the background.

### MWIR

3–5 µm band, detector ≈ −196 °C

### ~100 m

Target range from a ~10 m pole

### AI plume

Automatic segmentation, multi-frame confirmation

### GLD + OGI

Point and area in one dashboard

MWIR camera → Edge server → GLD + OGI dashboard

Next-stage development concept · target specifications · 3D illustrations

PERTAMINA PATRA NIAGA

PT Pertamina Patra Niaga (Reliability Function) · LAPI Ganesha Utama · Institut Teknologi Bandung